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LEAD DEVOPS/PLATFORM ENGINEER

LEAD DEVOPS/PLATFORM ENGINEER Experienced DevOps and Platform Engineer with over a decade of leadership in architecting and optimizing scalable Kubernetes and MLOps solutions across multi-cloud environments. Expert in automating infrastructure with Terraform, streamlining CI/CD with tools like GitHub Actions and ArgoCD, and securing microservices using Istio. Adept at deploying and managing ML workflows using Databricks, MLflow, and TinyML on Kubernetes. Proven ability to align infrastructure strategies with business goals to enhance agility, performance, and innovation in AI-driven environments..

TECHNICAL SKILLS

- **Cloud Computing:** AWS, GCP, Azure
- **Frameworks:** Agile Scrum, Jenkins, GitOps (ArgoCD), Helm, kubectl, Kustomize, Tilt
- **MLOps/AI/ML Tools:** TensorFlow, TensorFlow Lite, MLflow, ONNX, SageMaker, Azure ML, Google AI Platform, TinyML, scikit-learn, Jupyter, edX Harvard TinyML Suite (Application, Deployment, Deployment at Scale), Databricks (MLflow, Feature Store, Model Registry)
- **Versioning:** Git, GitHub, Bitbucket, GitHub Action, GitLab
- **Scripting:** Python, GoLang, Shell Scripts, Bash
- **Automation & Configuration:** Ansible, Terraform, Python,
- **Container Runtimes:** Docker, Containerd, CRI-O
- **Orchestration:** Kubernetes, Docker-Swarm, OpenShift, CloudFoundry, Ranchers
- **Observability:** Prometheus, Grafana, Thanos, Loki, Promtail, Jaeger, Kiali, Zipkin, Ksnif, Wireshark, Dynatrace, Datadog
- **Networking:** Service Mesh Implementation
- **Security:** Cluster Setup, Cluster Hardening, System Hardening, Monitoring Logging & Traceability.

CAREER HIGHLIGHTS

- Designed and managed scalable Kubernetes clusters, reducing deployment time by 30% for McKinsey & Company's AI solutions.
- Integrated TinyML inference pipelines using TensorFlow Lite in GCP-hosted Kubernetes, enabling real-time insights on edge devices.
- Orchestrated end-to-end MLOps workflows at Cigna using MLflow and Terraform, reducing ML model deployment time by 50%
- Implemented automated MLOps pipelines on Databricks for model training, evaluation, deployment, and monitoring across hybrid cloud infrastructure.
- Optimized Kubernetes configurations across multiple cloud platforms, reducing infrastructure costs by 20% for Cigna.
- Implemented Istio service mesh, improving microservices performance by 25% and reducing security vulnerabilities by 35%.
- Enhanced system monitoring and logging, improving issue detection and resolution times by 40%.
- Developed a custom Kubernetes abstraction layer and self-service platform, increasing developer productivity by 40% and reducing operations workload by 60%.
- Implemented GitOps practices and multi-cluster solutions, improving deployment consistency by 70% and resource utilization by 80%.
- Integrated advanced CI/CD pipelines and custom Kubernetes operators, reducing deployment times and manual interventions significantly.
- Developed a custom self-service MLOps and DevOps platform at McKinsey using GitOps and ArgoCD, boosting engineering throughput by 40%.

DEVELOPER EXPERIENCE

- **Advanced Implementation of Cloud-Native PostgreSQL (PG).** Successfully implemented cloud-native PostgreSQL databases to support scalable and resilient applications.
- **Impact:** Enhanced database performance and reliability, ensuring seamless integration with Kubernetes environments.
- **External Secret Operator for Effective Secret Management.** Utilized the External Secret Operator to manage sensitive information securely within Kubernetes clusters.
- **Impact:** Improved security by automating the synchronization of secrets from external secret management systems, reducing manual intervention and potential security risks.
- **TILT for Efficient Iteration Through Changes.** Leveraged TILT to streamline the development workflow by enabling rapid iteration and feedback loops.
- **Impact:** Accelerated development cycles by allowing developers to see the impact of their changes in real-time, reducing the time required for testing and debugging.
- **Kustomize for Environment-Specific Deployments.** Employed Kustomize to manage Kubernetes configurations across different environments, ensuring consistency and flexibility.
- **Impact:** Simplified the deployment process by enabling reusable and customizable configurations, reducing errors and improving deployment efficiency.
- **Kubectl for Facilitating Deployments Across Different Environments.** Used Kubectl to manage complex deployments across multiple environments, ensuring consistency and reliability.

- Impact: Enhanced deployment processes by providing a robust framework for managing and deploying Kubernetes resources, reducing deployment times and increasing reliability.
- Advanced Implementation of Istio Service Mesh. Successfully implemented Istio service mesh to enhance microservices architecture and communication within Kubernetes environments
- Impact: Enhanced microservices performance, security, and observability while enabling advanced deployment strategies.
- Ensured seamless integration with Kubernetes, providing a robust foundation for cloud-native applications.
- Built and deployed TinyML applications using TensorFlow Lite on Kubernetes-managed edge devices.
- Integrated TinyML inference and telemetry capture into GitOps-managed pipelines for continuous monitoring and updates.
- Architected scalable ML model deployment strategies using Terraform, MLflow, and cloud-native ML platforms (SageMaker, Azure ML, Databricks).
- Built production-ready MLOps workflows on Databricks, integrating MLflow for model tracking, Feature Store for feature versioning, and Model Registry for lifecycle management.
- Extended Istio configuration to optimize real-time ML prediction APIs, enabling adaptive traffic control and retry policies.
- Leveraged Kustomize and Kubectl for deploying versioned ML pipelines across dev, test, and production environments.

PROFESSIONAL EXPERIENCE

Manager/Lead Kubernetes Engineer I McKinsey & Company, Dallas, TX

Aug 2023 to Date

- Architected and deployed scalable Kubernetes clusters tailored for production and AI workloads, incorporating best practices for performance and reliability while ensuring seamless integration with existing systems and infrastructure.
- Automated the provisioning and management of Kubernetes clusters using Infrastructure as Code (IaC) tools such as Terraform, thereby improving consistency, repeatability, and speed of deployment across multiple environments.
- Built and maintained CI/CD pipelines for containerized applications using GitHub Actions, Helm, and Gitlab, streamlining the development process and ensuring rapid, reliable software delivery with minimal manual intervention.
- Monitored and optimized cluster performance with Prometheus/Grafana, Dynatrace, and native Kubernetes tools, implementing proactive measures to maintain high availability, secure multi-tenant environments, and zero downtime during upgrades.
- Set up and managed Istio service mesh to facilitate secure and efficient microservices communication, implementing advanced traffic management policies like load balancing, traffic splitting, and canary deployments to enhance application performance and reliability.
- Proficient in deploying and managing containerized applications using Google Kubernetes Engine (GKE) on Google Cloud Platform (GCP), including setup and configuration of Kubernetes clusters.
- Hands-on experience with Infrastructure as Code (IaC) using Terraform for automating GKE cluster provisioning, scaling, and maintenance.
- Managed cloud networking and connectivity for GKE clusters, including VPN configurations, VPCs, and secure communication between on-premises data centers and GCP environments.
- Implemented and maintained CI/CD pipelines for automated deployment of microservices on GKE, integrating tools like Jenkins, GitHub Actions, and Google Cloud Build.
- Experienced in monitoring and optimizing GKE cluster performance using Google Cloud's monitoring tools, Prometheus, and Grafana, ensuring high availability and reliability.
- Boosted application security by integrating Istio's mTLS, JWT authentication, and network policy enforcement while developing custom Istio operators to automate sidecar injection and streamline policy management for robust microservice security.
- Configured Istio ingress gateways for secure external access to microservices, monitored and analyzed microservice traffic using Istio's telemetry tools, and optimized the performance of Istio control and data planes to ensure smooth operation in production environments.
- Established and maintained API Gateways, significantly enhancing application security and performance while ensuring seamless integration and efficient traffic management for high-traffic websites and microservices architectures.
- Designed and deployed scalable load balancer solutions, utilizing them in conjunction with API gateways to support high-performance architecture while ensuring optimal resource utilization and failover capabilities and monitoring traffic distribution and API performance.
- Led deployment of Kubernetes-native MLOps pipelines, integrating MLflow and Databricks for automated model training, evaluation, and promotion to production.
- Designed scalable infrastructure to support ML applications and centralized training pipelines.
- Collaborated with AI teams to containerize models, apply GitOps workflows, and deploy across hybrid environments using GKE and Istio for secure inference routing.
- Built monitoring pipelines using Prometheus and Databricks metrics for ML performance tracking and retraining triggers.
- Deployed and monitored applications on OpenShift, ensuring efficient resource utilization and high availability.
- Utilized Cloud Foundry for streamlined application deployment and management across multiple environments.
- Automated deployment pipelines for Cloud Foundry applications, enhancing deployment speed and reliability.
- Implemented security best practices for containerized applications in both OpenShift and Cloud Foundry environment.
- Deployed and monitored applications on OpenShift, ensuring efficient resource utilization and high availability.
- Utilized Cloud Foundry for streamlined application deployment and management across multiple environments.
- Automated development for Cloud Foundry applications, enhancing deployment speed and reliability.
- Implemented security best practices for containerized applications in both OpenShift and Cloud Foundry environment.

Lead Kubernetes Engineer I Cigna, Bloomfield, CT

Feb 2022 to Aug

2023

- Configured and maintained Kubernetes clusters across various cloud platforms, including GKE, EKS, and AKS, ensuring optimal performance and reliability through regular updates and troubleshooting.
- Supported Kubernetes production operations, providing robust solutions for container hardening, remediation, and secure infrastructure provisioning using Terraform to maintain high standards of security and compliance.
- Installed, maintained, and provided high-availability support for HashiCorp Vault, trained developers on Kubernetes best practices, and facilitated the onboarding of applications to Kubernetes for seamless integration and deployment.
- Configured and managed Istio service mesh to enhance microservices communication, provisioned and maintained monitoring and logging applications like Prometheus and EFK, and streamlined application lifecycle management using Helm and ArgoCD.
- Installed and supported various Kubernetes cluster tools, including Vault, cert-manager, Dex, ingress controllers, and the EFK stack, to ensure comprehensive cluster management and security.
- Designed disaster recovery solutions for both cloud and on-premises environments, automated system buildouts and application deployment processes, and collaborated with development teams to engineer scalable, reliable, and resilient cloud-based software.
- Implemented ML model lifecycle automation using Databricks with Terraform provisioning and CI/CD orchestration across AKS and EKS clusters.
- Developed reusable templates for ML deployments across dev, QA, and production clusters, ensuring consistency and auditability.
- Collaborated with data science teams to onboard ML workflows into MLOps platforms and integrated model versioning via MLflow and Databricks Feature Store.
- Spearheaded observability enhancements using Grafana and Prometheus to monitor ML inference latency, feature drift, and model accuracy in real time.
- Deployed and monitored applications on OpenShift, ensuring efficient resource utilization and high availability.
- Utilized Cloud Foundry for streamlined application deployment and management across multiple environments.
- Automated deployment pipelines for Cloud Foundry applications, enhancing deployment speed and reliability.
- Implemented security best practices for containerized applications in both OpenShift and Cloud Foundry environment.
- Deployed and monitored applications on OpenShift, ensuring efficient resource utilization and high availability.
- Utilized Cloud Foundry for streamlined application deployment and management across multiple environments.
- Automated deployment pipelines for Cloud Foundry applications, enhancing deployment speed and reliability.
- Implemented security best practices for containerized applications in both OpenShift and Cloud Foundry environment.
- Developed a DevOps pipeline using Jenkins and AWS CI/CD tools, created tools to monitor cloud applications and services, wrote Ansible playbooks in YAML, and automated platform environment setup.
- Leveraged Terraform to architect and deploy serverless applications utilizing AWS Step Functions, Lambda, S3 for front-end hosting, and RDS Aurora Serverless for the backend while managing IAM Root, Users, and Groups in compliance with customer policies.
- Optimized cloud infrastructure by refining storage/compute/database resources, created and managed S3 Buckets with versioning, configured CloudFront as an HTTPS endpoint, and managed DNS records through AWS Route 53 for enhanced web application security.
- Configured secure web applications using Bastion Host, NAT Gateway, and Application Load Balancer within a custom VPC, established VPC peering for internal and external accounts, and integrated nightly builds with SonarQube and Veracode.
- Proficient in deploying and managing containerized applications using Google Kubernetes Engine (GKE) on Google Cloud Platform (GCP), including setup and configuration of Kubernetes clusters.
- Hands-on experience with Infrastructure as Code (IaC) using Terraform for automating GKE cluster provisioning, scaling, and maintenance.
- Managed cloud networking and connectivity for GKE clusters, including VPN configurations, VPCs, and secure communication between on-premises data centers and GCP environments.
- Implemented and maintained CI/CD pipelines for automated deployment of microservices on GKE, integrating tools like Jenkins, GitHub Actions, and Google Cloud Build.
- Experienced in monitoring and optimizing GKE cluster performance using Google Cloud's monitoring tools, Prometheus, and Grafana, ensuring high availability and reliability.

Lead Site Reliability Engineer (SRE) I DevOps Engineer I Wells Fargo, Dallas, TX

Oct 2021 to

Feb 2022

- Monitored and ensured the availability of services, proper functioning of the underlying infrastructure, and reliability of internal tools, processes, and systems while specifying Service Level Indicators and Objectives to maintain high performance and compliance.
- Managed incidents and disaster recovery, facilitated post-incident analysis for continuous improvement, documented procedures, developed and maintained software factoring, and secured and maintained infrastructural deployments.
- Deployed and monitored applications on OpenShift, ensuring efficient resource utilization and high availability.
- Utilized Cloud Foundry for streamlined application deployment and management across multiple environments.
- Automated deployment pipelines for Cloud Foundry applications, enhancing deployment speed and reliability.
- Implemented security best practices for containerized applications in both OpenShift and Cloud Foundry environment.

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- Implemented security best practices for containerized applications in both OpenShift and Cloud Foundry environment.
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- Hands-on experience with Infrastructure as Code (IaC) using Terraform for automating GKE cluster provisioning, scaling, and maintenance.
- Managed cloud networking and connectivity for GKE clusters, including VPN configurations, VPCs, and secure communication between on-premises data centers and GCP environments.
- Implemented and maintained CI/CD pipelines for automated deployment of microservices on GKE, integrating tools like Jenkins, GitHub Actions, and Google Cloud Build.
- Experienced in monitoring and optimizing GKE cluster performance using Google Cloud's monitoring tools, Prometheus, and Grafana, ensuring high availability and reliability.

Senior Site Reliability Engineer (SRE) | DevOps Engineer (Contract) | LogRhythm, Inc., Dallas, TX
to Oct 2021

Mar 2021

- Stayed up-to-date with evolving compliance mandates to ensure all systems and processes met regulatory standards and detected unauthenticated cross-account attacks on AWS to maintain robust security postures.
- Implemented Cyber Kill Chain steps and defenses against advanced persistent threats (APTs), utilizing LogRhythm tools for automation use cases in threat detection, mitigation, and reporting of cyber-attacks.
- Engaged in threat hunting and analysis, supported 24x7 mission-critical internet applications, managed QA and production deployments, documented new procedures, and provided support for both production and non-production application environments.
- Managed Linux projects by tracking application performance metrics, implementing cloud orchestration for end-user deployment, performing security analysis, and setting up Kubernetes for complex container orchestration and automated monitoring systems.
- Employed Terraform to construct multiple serverless applications using Step Functions, Lambda, and S3 for front-end hosting, with RDS Aurora Serverless as the back-end while managing IAM Root, Users, and Groups in line with customer compliance policies.
- Optimized cloud infrastructure to minimize costs related to storage, computing, and databases, managed S3 Buckets with versioning, configured CloudFront as an HTTPS endpoint, and managed DNS records to establish a secure and efficient networking system.
- Troubleshoot server issues, managed backup and recovery functions for PostgreSQL, and handled cascading and multiple replications of PostgreSQL 10 to ensure data integrity and high availability.
- Deployed and monitored applications on OpenShift, ensuring efficient resource utilization and high availability.
- Utilized Cloud Foundry for streamlined application deployment and management across multiple environments.
- Automated deployment pipelines for Cloud Foundry applications, enhancing deployment speed and reliability.
- Implemented security best practices for containerized applications in both OpenShift and Cloud Foundry environment.
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- Hands-on experience with Infrastructure as Code (IaC) using Terraform for automating GKE cluster provisioning, scaling, and maintenance.
- Managed cloud networking and connectivity for GKE clusters, including VPN configurations, VPCs, and secure communication between on-premises data centers and GCP environments.
- Implemented and maintained CI/CD pipelines for automated deployment of microservices on GKE, integrating tools like Jenkins, GitHub Actions, and Google Cloud Build.
- Experienced in monitoring and optimizing GKE cluster performance using Google Cloud's monitoring tools, Prometheus, and Grafana, ensuring high availability and reliability.

Senior Cloud, DevOps & Kubernetes Engineer | JP Associates, Columbia City, MD
to Mar 2021

Jul 2016

- Automated system buildouts and streamlined application deployment processes, collaborating closely with the development team to engineer scalable, reliable, and resilient cloud-based software solutions.
- Implemented a DevOps pipeline utilizing Jenkins and AWS CI/CD tools, ensuring efficient software delivery and continuous integration across development environments.
- Developed and utilized monitoring tools to track application and service performance in the cloud while writing YAML-based Ansible playbooks from scratch and managing platform environments through automated setups and troubleshooting.
- Implemented secure web applications using Bastion Host, NAT Gateway, and Application Load Balancer within a customized VPC and established VPC peering to facilitate access and routing across internal and external accounts.
- Orchestrated nightly builds integrated with code quality tools like SonarQube and Veracode, configured Kubernetes for complex containerization systems, and designed solutions leveraging Kubernetes clusters for optimized application deployment and scalability.

- Proficient in deploying and managing containerized applications using Google Kubernetes Engine (GKE) on Google Cloud Platform (GCP), including setup and configuration of Kubernetes clusters.
- Hands-on experience with Infrastructure as Code (IaC) using Terraform for automating GKE cluster provisioning, scaling, and maintenance.
- Managed cloud networking and connectivity for GKE clusters, including VPN configurations, VPCs, and secure communication between on-premises data centers and GCP environments.
- Implemented and maintained CI/CD pipelines for automated deployment of microservices on GKE, integrating tools like Jenkins, GitHub Actions, and Google Cloud Build.
- Experienced in monitoring and optimizing GKE cluster performance using Google Cloud's monitoring tools, Prometheus, and Grafana, ensuring high availability and reliability.
- Managed peripherals, services, settings, and storage configurations, deployed cloud-native applications, monitored and supported Kubernetes environments, and demonstrated expertise in on-premise to cloud migrations.
- Deployed and monitored applications on OpenShift, ensuring efficient resource utilization and high availability.
- Utilized Cloud Foundry for streamlined application deployment and management across multiple environments.
- Automated deployment pipelines for Cloud Foundry applications, enhancing deployment speed and reliability.
- Implemented security best practices for containerized applications in both OpenShift and Cloud Foundry environment.
- Deployed and monitored applications on OpenShift, ensuring efficient resource utilization and high availability.
- Utilized Cloud Foundry for streamlined application deployment and management across multiple environments.
- Automated deployment pipelines for Cloud Foundry applications, enhancing deployment speed and reliability.
- Implemented security best practices for containerized applications in both OpenShift and Cloud Foundry environment.

ADDITIONAL EXPERIENCE

Cloud & Linux Engineer | Zhill Systems, Dallas, TX

Linux Administrator | Aived International Services, Lagos, Nigeria

EDUCATION AND CERTIFICATIONS

Post Graduate Program in Artificial Intelligence and Machine Learning: Business Applications | University of Texas

Bachelor of Science (BS) in Biochemistry | Usmanu Dan Fodiyo University, Sokoto, Nigeria

TinyML Application, Deployment, and Deployment at Scale | edX/Harvard University

Certified AWS Solution Architect | Amazon Web Services (AWS)

Certified Kubernetes Administrator (CKA) | Linux Foundation